SOVEREIGN AI WORKBENCH — INDUSTRIAL INSPECTION REPORT

Refinery Unit 4B — Operational Safety Assessment 2026

Target Equipment: Boiler Turbine Generator BTG-404 | Date: 2026-08-29

1. EXECUTIVE SUMMARY & INSPECTION FINDINGS

During routine thermal and pressure audit of Boiler Turbine Generator BTG-404:

- Primary Steam Temperature registered at 565.0 Degrees C.
- Main Steam Header Pressure registered at 58.5 BAR.
- Safety SOP Limit for Main Steam Header Pressure: 52.0 BAR.
- Calculated Exceedance: +6.5 BAR (+12.5% over maximum allowable threshold).

2. MANDATORY OPERATIONAL SAFETY PROTOCOL

Section 8.4: Main Steam Overpressure Response Protocol.

When main steam header pressure exceeds 52.0 BAR for more than 120 seconds:

1. Immediately throttle main steam control valve MSV-1.
2. Initiate automated load reduction to 60% capacity.
3. Issue Form-B Emergency Governance Approval Note for senior engineer sign-off.